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# Interventions addressing the adolescent HIV continuum of care in South Africa: a systematic review and Delphi analysis



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# Interventions addressing the adolescent HIV continuum of care in South Africa: a systematic review and Delphi analysis

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What is already known about this subject?

- Adolescents have significant gaps along the continuum of HIV care
- Few interventions have successfully improved clinical care for adolescents in low- and middle-income countries

What are the new findings?

- Use of alternative venues including home HIV testing have promise in increasing HIV diagnosis among adolescents
- Currently there are no proven effective interventions to successfully link adolescents living with HIV to care in South Africa
- Non-traditional healthcare delivery systems including weekend clinics and home assessments have potential to increase engagement in care among adolescents
- Currently there are no interventions addressing transition from pediatric to adult care for adolescents living with HIV in South Africa

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- There are limited cost analyses for interventions addressing the adolescent HIV continuum of care in South Africa

What are the recommendations for policy and practice?

- Future interventions should work comprehensively across the adolescent HIV continuum of care
- Future interventions should address costs of implementation
- Future interventions should be tailored to the specific needs of adolescents

## Strengths and Limitations of the Study

- The AHI(SA)$^2$ network includes highly experienced experts with key knowledge to evaluate interventions in South Africa.
- We evaluate interventions with quantifiable results addressing the continuum of care for adolescents and young adults living with HIV evaluated by scholars with expertise in this population.
- The evaluation included interventions that were published in the literature and did not include studies that were in progress or planned.
- Not all AHI(SA)$^2$ members who were invited to participate completed the survey

**Abstract:**

**Introduction:** Compared to adults, adolescents in South Africa have larger gaps at each step of the HIV continuum of care resulting in low levels of viral suppression.

**Methods:** We conducted a systematic review and Delphi analysis of interventions addressing the HIV continuum of care for adolescents in South Africa. We searched PubMed, Science Direct, and Google Scholar and online conference proceedings from the International AIDS Society, the International AIDS Conference, and the Conference on Retrovirology and Opportunistic Infections from January 1, 2010 to September 30, 2020. We then conducted a Delphi analysis with 29 researchers involved in the National Institutes of Health's Fogarty International supported Adolescent HIV Implementation Science Alliance – South Africa to evaluate interventions for efficacy, feasibility, and potential for scale-up.

**Results:** We identified nine initial published articles containing interventions addressing the adolescent HIV continuum of care in South Africa, including five interventions focused on HIV diagnosis, two on antiretroviral therapy adherence, and two on retention in care. No studies addressed linkage to care or transition from pediatric to adult care. Two studies discussed intervention costs. In-home and HIV-self testing, community-based adherence support, and



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provision of adolescent-friendly services were the most impactful and scalable interventions addressing the adolescent HIV continuum of care.

**Conclusion:** Future interventions should work comprehensively across the adolescent HIV continuum of care and be tailored to the specific needs of adolescents.

**Introduction:**

South Africa has the highest burden of adolescents and young adults living with HIV (ALWH) in the world.[1] In the late 1990s and early 2000s, delays in access to HIV treatment and suboptimal HIV prevention tools contributed to more than 330,000 children being born with HIV through perinatal transmission.[2-5] These children are now aging into adolescence and young adulthood. In addition, non-perinatal (typically sexual) transmission is a major concern among adolescents in South Africa. In 2017, an estimated 50,200 adolescents aged 15-19 contracted HIV, with the number of females (n=41,000) being more than four times higher than males (n=9,200).[1] Despite growing recognition of the challenges in adolescent HIV prevention and care, HIV/AIDS remains the leading cause of death among all adolescents in South Africa.[6]

South Africa has the largest antiretroviral therapy (ART) program in the world, and national ART guidelines support early initiation of ART with simple, potent, and well-tolerated regimens.[7] ART scale-up has contributed to high levels of viral suppression, low mortality rates, and low HIV transmission among adults living with HIV in South Africa.[1, 8, 9] However, the continuum of care for ALWH (i.e., from diagnosis to linkage to care and subsequent ART adherence, retention in care and viral suppression) has significant gaps at each step, resulting in only 10% of ALWH in South Africa being virally suppressed in a study conducted in 2016.[2] In addition, the process of transition from pediatric to adult care lacks evidence-based guidelines and has shown poor levels of retention in care and viral suppression after transition.[10-12]

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While research is ongoing to support HIV treatment for ALWH in South Africa, the extent and efficacy of interventions has not been well defined, and evidence-based guidance is needed to direct future intervention development. Our objective was to utilize a network of researchers working to improve HIV care for ALWH in South Africa to highlight and expand implementation of successful interventions for this population. Here, we present the results of a systematic review and Delphi analysis of interventions addressing the adolescent HIV continuum of care in South Africa.

## Methods

*Systematic review*

We searched PubMed, Science Direct, and Google Scholar and online conference proceedings from the International AIDS Society (IAS), the International AIDS Conference (AIDS), and the Conference on Retrovirology and Opportunistic Infections (CROI) from January 1, 2010 to September 30, 2020. Key words and medical subject headings relevant to age (i.e., adolescent, adolescence, teen, young adult) were cross-referenced with terms associated with the HIV continuum of care (i.e., diagnosis, linkage to care, retention, adherence, and transition). We reviewed literature published in English that reported interventions addressing the adolescent and young adult HIV continuum of care that were piloted or implemented in South Africa among adolescents and young adults. We adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[13] Because of variability in reported age ranges, we included studies in our analysis with primary data on youth ages 10 to 24 years. We present age ranges as described by the original authors.

All identified studies were uploaded into Covidence, a non-profit website working with the Cochrane database to improve systematic reviews (www.covidence.org, Melbourne, Australia). Duplicate materials were removed. After initially screening the titles and abstracts, two authors (BCZ and TR) independently reviewed potential studies. Conflicts were resolved by reviewing the



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full text article and discussing the inclusion/exclusion criteria until consensus was reached. We excluded review articles and studies that did not include an intervention, did not report primary data affecting the HIV continuum of care, or were targeting children or adults outside our specified age range of 10 to 24 years. We then extracted data from the full text articles and created summarized tables organized by each step in the HIV continuum of care.

We used the PRISMA guidelines to assess the strength of evidence and bias for all included clinical trials and evaluated random sequence generation, allocation concealment, blinding, and outcome reporting.[13, 14] Randomized controlled trials were considered less biased than pre-post evaluations, and prospective and retrospective observational studies.

*Delphi analysis*

To identify the most effective, feasible and scalable interventions impacting the adolescent HIV continuum of care in South Africa, we engaged the National Institute of Health's Fogarty International supported Adolescent HIV Implementation Science Alliance – South Africa (AHI(SA)$^2$ (www.ahisa2.org). Members of AHI(SA)$^2$ were selected by inviting (via email) researchers with current NIH funding and/or published first authors (and their research teams) addressing adolescent HIV prevention or the adolescent HIV continuum of care in South Africa. We uploaded summarized tables from our systematic review, along with the full text articles and abstracts, into REDCap.[15] We invited AHI(SA)$^2$ members to complete a questionnaire assessing the feasibility, efficacy, scalability, and costing of each intervention (using four-point Likert scales). We also asked members to assign an overall score (rating of 0 to 100) regarding the potential impact of the intervention on the continuum of care for ALWH in South Africa.

All AHI(SA)$^2$ members (n=42) received a secure link to the REDCap survey via email to complete the first-round questionnaire focused on intervention efficacy (August 24, 2020 – September 23,



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2020). Results were discussed at the first AHI(SA)$^2$ virtual meeting on September 25, 2020 with the 25 participants who attended. Interventions with a prespecified median overall impact score less than 70 or with an average efficacy score less than three (out of four on the Likert scale) were removed from the next round questionnaire. In preparation for the second round, we updated the literature search from July 31, 2020 to September 30, 2020 and invited AHI(SA)$^2$ members to nominate additional studies for inclusion. The REDCap questionnaire was updated and emailed to AHI(SA)$^2$ members who participated in the first-round questionnaire (n=29) to evaluate feasibility and scale (September 25, 2020 to December 15, 2020). We sought to formulate an expert consensus on the interventions for ALWH with highest potential be taken to scale and to highlight areas in which future interventions are needed.[16]

*Patient and public involvement*

Patients and/or the public were not involved in the design or conduct of this research. The research question and design were discussed with the research team at the initial AHI(SA)$^2$ meeting. Results of each round of the Delphi analysis were discussed at AHI(SA)$^2$ virtual meetings.

**Results**

*Systematic review*

Our initial search yielded 273 relevant articles. Of these, 110 were removed due to duplicate publications or interventions, and 112 did not include interventions addressing the adolescent HIV continuum of care in South Africa, leaving 25 full text articles for review. We excluded 16 of the 25 studies because nine lacked an intervention, three focused on HIV primary prevention, and four met one of the following exclusionary criteria: literature review only (n=1), no primary results (n=1), not focused on adolescents (n=1), or not conducted in South Africa (n=1). We did not find any additional studies in the review of conference abstracts. Overall, nine published papers with

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sufficient data were included in the first round of the Delphi analysis addressing the adolescent and young adult HIV continuum of care in South Africa.

*Delphi analysis*

Participants: AHI(SA)[2] consisted of 42 eligible individuals involved in adolescent HIV research in South Africa. For the first round of the modified Delphi analysis, 29 (69%) participated in the REDCap survey. Participant characteristics are presented in **Table 1**. Of the 29 who initiated the survey, 21 (72%) completed it. Of the 29 participants who were sent the second-round survey, 21 (72%) initiated it and 18 (62%) completed the survey.

**Table 1: Participant characteristics in two rounds of the modified Delphi Analysis.**

| Demographics | First Round (N = 29) | Second Round (N=21) |
|---|---|---|
| | n (%) | n (%) |
| **Profession** | | |
| Researcher | 13 (45%) | 12 (57%) |
| Clinician/researcher | 6 (21%) | 4 (19%) |
| Medical doctor | 6 (21%) | 3 (14%) |
| Healthcare provider | 2 (7%) | 1 (5%) |
| Student/fellow | 2 (7%) | 1 (5%) |
| **Years working in adolescent HIV** | | |
| > 10 years | 14 (48%) | 10 (48%) |
| 6 – 10 years | 6 (21%) | 6 (29%) |
| 1 – 5years | 8 (28%) | 4 (19%) |
| < 1 year | 1 (3%) | 1 (5%) |
| **Highest Educational Degree** | | |
| PhD | 11 (38%) | 9 (43%) |
| MD or MBChB | 5 (17%) | 2 (10%) |
| MPH or other master's degree | 5 (17%) | 4 (19%) |
| MD/MBChB + PhD | 3 (10%) | 3 (14%) |
| MD/MBChB + master's degree | 2 (7%) | 3 (14%) |
| Other | 3 (10%) | --- |
| **Primary residence** | | |
| South Africa | 22 (76%) | 15 (71%) |
| United States | 6 (21%) | 4 (19%) |
| United Kingdom | 1 (3%) | 1 (5%) |

Publications:



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Among the nine initial published articles containing interventions addressing the adolescent HIV continuum of care in South Africa, five interventions focused on HIV diagnosis, two on ART adherence, and two on retention in care as indicated in **Table 2**. No studies addressed linkage to care or transition from pediatric to adult care. For study design, five studies were randomized controlled trials, three were retrospective cohort analyses, and one was a pre-post evaluation. Two studies of the nine discussed costing of the intervention.

Intervention evaluation:

Below we report summaries of the individual published interventions for ALWH in South Africa separated by steps in the continuum of care. Within each step, the interventions are listed beginning with the highest overall median impact rating in the Delphi Analysis (**Tables 2 and 3**).

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**Table 2:** Interventions to improve the continuum of care for adolescents and young adults living with HIV in South Africa

| Authors | Title | Publication and Date | Location | Number of Subjects | Ages (Target Population) | Study Type | Intervention Description | Overall Results | Median Score Round 1 |
|---|---|---|---|---|---|---|---|---|---|
| **Adolescent HIV Testing** | | | | | | | | | |
| Doherty, et al. | Effect of Home-Based HIV Counseling and Testing Intervention in Rural South Africa: Cluster Randomized Trial | *BMJ* 2013 | uMzimkhulu subdistrict, KwaZulu-Natal, South Africa | 4154 Adolescents and Adults | 14-17, 18+ Adult Guardians | Cluster Randomized Controlled Trial; Door-to-Door Testing | Door-to-door home-based HIV testing (HBT) performed by lay counselors compared to standard of care HIV counseling and testing at local clinics. | Testing completed in 69% of HBT compared to 47% in control (PR 1.54 95% CI 1.32 to 1.81) | 72.5 |
| Rotheram-Borus, et al. | Feasibility of Using Soccer and Job Training to Prevent Drug Abuse and HIV | *AIDS and Behavior* 2016 | South Africa | 142 Adolescents | 18-25 Unemployed Males | Community Randomized Controlled Trial | Randomized neighborhoods to immediate intervention or delayed control; Intervention: Grassroot Soccer program with trained coaches, random rapid diagnostic tests for alcohol/drugs, and vocational training. | 29% testing in immediate vs. 24% in delayed (no significant difference) | 50 |
| Hershow, et al. | Using Soccer to Build Confidence and Increase HCT Uptake Among Adolescent Girls | *Sport in Society* 2015 | South Africa | 1953 Adolescents | 12-16 (Females Only) | Pre-Post Evaluation | Survey before and after participation in SKILLZ Street intervention developed by Grassroot Soccer (GRS). SKILLZ Street female coaches delivered an afterschool education program consisting of 10, two hour bi-weekly sessions. | 69% tested for HIV<br>No control group. | 63 |
| Shanaube, et al. | Community Intervention Improves Knowledge of HIV Status of Adolescents in Zambia: Findings from HPTN 071-PopART for Youth Study | *AIDS* 2017 | Results from Zambia only; Intervention also conducted in South Africa | 11 175 Adolescents | 15-19 | Community-Randomized Clinical Trial | **HPTN 071 (PopART)**<br>Door-to-door HIV prevention package delivered by community health worders. Intervention included in-home HIV Counseling and Testing (HCT), support for linkage to care for individuals identified living with HIV, ART adherence support, referrals for voluntary medical male circumcision | HCT uptake 80% (8,707/10,809) and awareness of an HIV-positive status tripled from 75 to 210. Home visits from providers saw the percentage of ALWH aware of their HIV status rise from 27.6% to 88.5%.<br><br>No comparison with standard care. | 76 |
| Pettifor, et al | HIV self-testing among young women in rural South Africa: A randomized controlled trial comparing clinic-based HIV testing to the choice of either clinic | *EClinical Medicine* 2020 | Mpumlanga, South Africa | 287 women | 18 - 26 | Randomized Controlled Trial | Standard of care HIV counseling and testing (HCT) at government clinics vs. choice of clinic based HCT or HIV self-testing kits.<br><br>+referrals for peers to receive clinic based HCT or HIV self-testing kits | 146 randomized to HCT and 141 to choice. In the choice arm 95% (n=135) chose HIV self testing kits. After 3 months a significantly higher percent (92%) of participants in the choice arm reported HIV testing compared to (43%) the HCT arm with a difference of 43% (95% CI 40% - 58%). By 9 months the difference decreased to 25% (95% CI 17% - 33%). Participants in the choice | 70 |



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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | testing or HIV self-testing with secondary distribution to peers and partners | | | | | | | arm were more likely to invite peers or partners to test (94% vs. 76%) | |
| **Adherence** | | | | | | | | | |
| Bhana, et al. | The VUKA Family Program: Piloting a Family-Based Psychosocial Intervention to Promote Health and Mental Health Among HIV Infected Early Adolescents in South Africa | *AIDS Care* Jan 2014 | KwaZulu-Natal, South Africa | 65 Adolescents and Their Family Members | 10-13 | Randomized Controlled Trial | **VUKA Family Program** Inspired by CHAMP, the intervention consisted of 3 months and ten sessions involving a culturally-tailored cartoon storyline and curriculum. The curriculum follows the story and provides step-by-step guidance for problem solving and facilitating discussions. ALWH were assessed for feelings of stigma, knowledge of ART, adherence depression and caregiver support. | ART adherence: Intervention (I) 1.10, Control (C) -. 43 at baseline; Follow-up, $p < .05$.<br><br>Caregiver/child communication: (I) .41, (C) -.07, $p < .09$.<br><br>Youth knowledge of HIV treatment: (I) .27; (C) -.74, $p < .08$. | 68.5 |
| Cluver, et al. | Achieving Equity in HIV-Treatment Outcomes: Can Social Protection Improve Adolescent ART-Adherence in South Africa? | *AIDS Care* 2016 | Eastern Cape, South Africa | 1059 Adolescents | 10-19 | Observational Cohort | Evaluated the provision of social protections on adherence. Social provisions included government cash transfers, food security, school fees/materials, school feeding, and clothing; HIV support group, sports groups, choir/art groups, positive parenting and parental supervision/monitoring. | Three social protection provisions were associated with reduced non-adherence:<br><br>Food provision (aOR .57, CI .42–.76, p < .001); HIV support group attendance (aOR .60, CI .40–.91, p < .02); High parental/caregiver supervision (aOR .56, CI .43–.73, p < .001). | 75 |
| **Retention in Care** | | | | | | | | | |
| Fatti, et al. | The Effectiveness and Cost-Effectiveness of Community-Based Support for Adolescents Receiving Antiretroviral Treatment: An Operational Research Study in South Africa | *Journal of the International AIDS Society* 2018 | KwaZulu-Natal/Western Cape/Eastern Cape/Mpumalanga, South Africa | 6706 Adolescents | 10-24 | Retrospective Cohort Study | Community based support (CBS) included home-based ART-related education, psychosocial support, symptom screening for opportunistic infections and support to access government grants. CBS was compared to standard of care. | Cumulative LTFU was 40% lower amongst participants receiving CBS (29.9%) compared to participants without CBS (38.9%), aHR = 0.60 (95% CI: 0.51 to 0.71); p < 0.0001).<br><br>Viral failure: at 3 years – no difference; at five years 18.8% CBS participants versus 37.2% non-CBS participants had viral failure, adjusted odds ratio = 0.24 (95% CI: 0.06 to 1.03). | 80 |
| Zanoni, et al. | Higher Retention and Viral Suppression with Adolescent-Focused HIV Clinic in South Africa | *PLoS One* 2017 | KwaZulu-Natal, South Africa | 254 HIV+ Adolescents | 13-24 | Retrospective Cohort Analysis | Weekend adolescent-focused clinic with peer support, group activities, counseling, and meals compared to standard of care. | Retention: 95% in adolescent-friendly clinic vs. 85% in standard clinic (OR 3.5; 95% CI 1.2 – 11.1; p=0.018).<br><br>Viral Suppression: 91% adolescent-friendly clinic vs. 80% in standard clinic. (OR 2.5; 95% CI 1.1 – 5.8; p=0.028 | 88 |

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**Table 3:** Second round Delphi analysis of interventions to improve the continuum of care for adolescents living with HIV in South Africa

| Authors | Title | Publication and date | Step in continuum of care | Costing | Feasibility mean score | Efficacy mean score | Scalability mean score | Overall median impact score |
|---|---|---|---|---|---|---|---|---|
| Zanoni, et al. | Higher Retention and Viral Suppression with Adolescent-Focused HIV Clinic in South Africa | *PLoS One* 2017 | Retention | $1.25 per adolescent per clinic visit | 3.34 | 3.28 | 3.17 | 80 |
| Fatti, et al. | The Effectiveness and Cost-Effectiveness of Community-Based Support for Adolescents Receiving Antiretroviral Treatment: An Operational Research Study in South Africa | *Journal of the International AIDS Society* 2018 | Retention | Cost of intervention was US$49.50/ patient/year. Incremental cost per patient-loss averted was US$600 and US$776 after 1 and 2 years, respectively. | 3.00 | 2.83 | 3.11 | 76 |
| Shanaube, et al. | Community Intervention Improves Knowledge of HIV Status of Adolescents in Zambia: Findings from HPTN 071-PopART for Youth Study | *AIDS* 2017 | HIV Diagnosis | Not evaluated | 3.39 | 3.11 | 3.00 | 75 |
| Pettifor, et al | HIV Self-testing among Young Women in Rural South Africa: A randomized controlled trial comparing clinic-based HIV testing to the choice of either clinic testing or HIV self-testing with secondary distribution to peers and partners | *EClinical Medicine* 2020 | HIV Diagnosis | Not evaluated | 3.06 | 2.76 | 3.00 | 72 |
| Cluver, et al. | Achieving Equity in HIV-Treatment Outcomes: Can Social Protection Improve Adolescent ART-Adherence in South Africa? | *AIDS Care* 2016 | ART Adherence | Not evaluated | 3.00 | 2.94 | 2.67 | 68 |
| Doherty, et al. | Effect of Home-Based HIV Counseling and Testing Intervention in Rural South Africa: Cluster Randomized Trial | *BMJ* 2013 | HIV Diagnosis | Not evaluated | 3.17 | 3.06 | 2.94 | 65 |

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HIV Diagnosis (n=5)

Five interventions addressed HIV diagnosis through expanded HIV testing among adolescents. Four of these were randomized controlled trials [17-20] and one was a pre-post evaluation.[21] Three of these studies met criteria for inclusion in both rounds of the Delphi analysis.[17, 18, 20] Two of the



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three studies showed high rates of HIV testing uptake and acceptability of in-home HIV testing and one showed strong short-term effects.

Shanaube et al. reported on the 11,175 adolescents randomized to receive the full intervention of door-to-door HIV testing and counseling with support for linkage to care delivered by community health workers compared to standard care in the Zambia PopART for Youth study.[20] HIV testing was done for 80% of participants with awareness of a positive HIV test increased by a three-fold by the third round of interventions and a median time to ART linkage of five months; however, there was no reported comparison to standard of care for HIV testing. Doherty et al. performed a cluster randomized clinic trial among adolescents evaluating door-to-door home-based HIV testing performed by lay counselors compared to standard of care.[17] HIV testing was significantly higher among in adolescents receiving home-based testing (69%) compared to standard of care (47%) (prevalence ratio 1.54, 95% CI 1.32 to 1.81). Pettifor et al. showed high uptake of HIV self-testing among young women. This study randomized 287 young women ages 18 to 26 to choose between HIV self-testing kits or receive clinic-based HIV testing compared to HIV testing in government clinics.[18] In the choice arm, 95% chose self-testing. After three months, 92% of the choice arm and 43% of the standard of care arm reported HIV testing a difference of 49%. By nine months, the difference reduced to 25%.

The remaining two studies did not meet the threshold to continue to the second round of the Delphi analysis due to lack of demonstrated efficacy as determined by the first round of the Delphi analysis; both used soccer as a means to increase HIV testing. Hershow et al. conducted a pre-post analysis of the SKILLZ Street intervention by Grassroot Soccer where female coaches delivered afterschool education program consisting of 10 two-hour bi-weekly sessions with 1,953 female adolescents ages 12 to 16 years.[21] Among the participants, 69% tested for HIV by the end of the intervention. There was no control group or comparison to baseline testing. Rotheram-Borus et al.



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randomized 142 male adolescents between the ages of 18 and 25 to immediately participate in Grassroot Soccer with trained coaches, rapid HIV testing, and vocational training compared to delayed participation in the control group.[19] There was no significant difference in HIV testing among the immediate group (29%) compared to the delayed uptake group (24%) after the intervention.

ART adherence (n = 2)

Two studies evaluated interventions designed to improve adolescent ART adherence.[22, 23] Both were included in first round of the Delphi analysis, but only one continued to the second round due to lack of perceived efficacy. Cluver et al. evaluated the provision of social protections on adherence among 1,059 adolescents ages 10 to 19 years in an observational cohort.[23] They found that food provision (adjusted odds ratio [aOR] 0.57, 95% CI 0.42 – 0.76), attendance at HIV support groups (aOR 0.60, 95% CI 0.40 – 0.91) and parental supervision (aOR 0.46, 95% CI 0.43 – 0.73) were associated with lower rates of self-reported non -adherence to ART in the prior week. Bhana et al.'s VUKA Family Program was not included in the second round of the Delphi analysis due to lack of clear efficacy. The intervention consisted of 10 sessions over three months and used a culturally tailored cartoon storyline of step-by-step guidance for problem solving skills and facilitated family discussions.[22] ART adherence (last time missed medication) increased over time from baseline to completion of the study, but there was no comparison to standard of care.

Retention in care (n=2)

Two studies were designed to improve retention in care, and both were included in the first and second rounds of the Delphi Analysis. Zanoni et al. evaluated a weekend-based, adolescent-focused clinic that offered peer support, group activities, counseling, and meals compared to standard of care in a retrospective cohort analysis.[24] Retention in care and viral suppression were higher among adolescents who attended the adolescent-friendly clinic (95% and 91%, respectively)



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compared to standard of care (85% and 80%, respectively; aOR 3.5, 95% CI 1.2 – 11.1; p=0.018 and aOR 2.5, 95% CI 1.1 – 5.8; p=0.028, respectively). The second study by Fatti et al. evaluated community-based support that included home-based ART education, psychosocial support, opportunistic infection screening, and access to government grants compared to standard of care among 6,706 adolescents aged 10 to 24 years.[25] Cumulative loss to follow up was lower in the intervention group (30%) compared to standard of care (39%; aHR = 0.60; 95% CI 0.51 – 0.71; p <0.0001), although there was no difference in viral failure after three or five years; 8.8% in the intervention group had viral failure compared to 37.2% in the standard of care group (aOR 0.24, 95%CI 0.06 – 1.03).

**Discussion:**

In our systematic review and Delphi analysis of interventions addressing the adolescent HIV continuum of care, we found five feasible and effective interventions with high potential for scale-up to improve adolescent HIV outcomes in South Africa. To improve HIV diagnosis among ALWH in South Africa, in-home HIV testing [18] and HIV self-testing [20] were promising interventions. However, there was a lack of demonstrated efficacy in targeting successful linkage to care after HIV testing. Adherence to ART among adolescents improved with the provision of food support, attendance in HIV support groups, and parental supervision.[23] Two interventions improved retention in care: one focused on adolescent-friendly services by providing an after-school hours clinic with peer support, counseling, and meals [24] and the other by providing in-home support for HIV education, psychosocial support, and assistance with applications to governmental grants.[25] Unfortunately, no interventions addressed transition from pediatric to adult based care for ALWH in South Africa.

Although adolescents and young adults have the highest incidence of new HIV infections in South Africa, they often do not present to health centers for HIV testing. Structural barriers such as school,



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transportation, or finances; psychosocial barriers such as the feelings of invincibility; or clinical factors such as the lack of clinical symptoms all contribute to poor engagement in care.[1, 6] Therefore, traditional methods including clinic-based opt-out testing are not ideal for this population. Non-traditional methods of intensified case finding are required to identify adolescents living with undiagnosed HIV. In-home HIV testing offers a solution to the issues of adolescents not attending health centers. Although this process can be labor intensive and costly, it also offers a mechanism to improve linkage to care once adolescents are diagnosed with HIV. Another potential solution to low HIV testing rates is self-testing which has shown acceptability and uptake but requires interventions to provide counseling to newly diagnosed individuals and facilitate linkage to care.

ALWH commonly suffer poor adherence to ART.[26, 27] Direct interventions addressing adherence among ALWH are often require considerable effort and resources and have variable efficacy.[28, 29] Indirect measures that improve food and economic security, psychosocial support, and supervision appear to improve adherence among adolescents.[23] Future interventions to address adherence should include economic and psychosocial support. In addition, addressing barriers such as depression and substance use which has improved ART adherence in adults may also benefit adolescents and young adults.[30]

After-school hours clinics, peer support, and improved connection to clinical staff have been shown to improve retention in care among ALWH.[31] As recommended by the World Health Organization, the provision of adolescent-friendly services and reimagining of clinical care delivery for adolescents and young adults can strengthen retention.[32] For example, many of these services can be provided outside of clinic space and time. In-home or virtual clinical services could be expanded to improve adolescent retention in care.



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Interventions require resources in time, personnel, and funding but only two interventions in this review presented costing analyses. For implementors and policymakers to adequately assess intervention scale-up, costing and cost-effectiveness analyses are required. Future interventions addressing the adolescent HIV continuum of care should evaluate cost.

We identified several interventions that could be scaled to improve the adolescent HIV continuum of care in South Africa; however, several gaps remain. Improving HIV diagnosis is critical to improve the continuum of care, but without linkage to care, adolescents cannot access ART and achieve viral suppression. Interventions to improve adolescent HIV testing should include linkage to care, and adolescents require additional support to achieve and maintain viral suppression. Moreover, children initiating ART under the care of pediatricians will eventually require transition to adult care; however, we did not find any published interventions addressing transition care for ALWH in South Africa highlighting a key area for future research.

This systematic review and Delphi analysis has several limitations. First, we only identified interventions that were published in the literature and did not include studies that were in progress or planned. In addition, the Delphi analysis did not include researchers who were not yet published, received non-NIH funding, or did not respond to email invitations. However, the AHI(SA)$^2$ network includes highly experienced experts with key knowledge to evaluate interventions in South Africa. In addition, not all AHI(SA)$^2$ members who were invited to participate completed the survey. However, this review highlights published interventions with quantifiable results addressing the continuum of care for adolescents and young adults evaluated by scholars with interest in this population.

**Conclusion:**



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No single intervention exists to improve each step of the continuum of care for ALWH in South Africa. We highlight several promising interventions, specifically in-home HIV testing, self-testing and alternative care delivery models. These interventions should be further evaluated as they are implemented to improve adolescent outcomes in the HIV care cascade. Future interventions should work comprehensively across the adolescent HIV continuum of care and be tailored to the specific needs of adolescents.

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**Declarations:**

**Author's Contributions:**
Dr. Zanoni conceptualized and designed the study, performed the literature review, conducted the analysis, drafted the initial manuscript, reviewed and revised the manuscript and approved the final manuscript as submitted.

Ms. Ramos performed the literature review, assisted with the analysis, assisted in the drafting of the initial manuscript, reviewed and revised the manuscript and approved the final manuscript as submitted.

Dr. Archary assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Ms. Sibaya assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

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Dr. Celum assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

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Dr. Bekker assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Haberer assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

**Competing Interests**: The authors have no conflicts of interest relevant to this article to disclose.



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**Ethics:** The study was given an exemption by the Emory University Institutional Research Ethics Board therefore it was not given a reference number. On review the committee felt that this was not Human Subjects Research but consensus generating.

**Availability of data:** The datasets used during the current study are available from the corresponding author on reasonable request.

**Acknowledgements:** Not Applicable

**References:**

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | Title |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | Abstract |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | p.4 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | p.4 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | p. 4 - 5 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | p. 4- 5 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | p. 4- 5 |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | p. 5 |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | p. 5 |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | p. 5 |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | p. 5 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | p. 5 |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | p. 5 |
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | p. 5 |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | n/a |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | p. 5 and 8 |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | n/a |
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | n/a |
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | n/a |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | p. 5 |
| Certainty assessment | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | n/a |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | p. 7 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | p. 7 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | p. 9 - 12 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | p. 9 - 12 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Table 2 |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | Table 2 |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | n/a |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | p. 9- 12 |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | n/a |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | n/a |
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | p. 9 -12 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | p. 12 - 14 |
| | 23b | Discuss any limitations of the evidence included in the review. | p. 14 |
| | 23c | Discuss any limitations of the review processes used. | p. 14 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | p. 12- 14 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | Not registered |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | AHISA documents |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | n/a |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | Title page |
| Competing interests | 26 | Declare any competing interests of review authors. | Title page |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | n/a |

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*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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# Interventions addressing the adolescent HIV continuum of care in South Africa: a systematic review and modified Delphi analysis

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# Interventions addressing the adolescent HIV continuum of care in South Africa: a systematic review and modified Delphi analysis

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**Word Count: 3355**

**Key Words:** Adolescent, HIV, South Africa, Intervention, Continuum of Care

**Abstract:**

**Introduction:** Compared to adults, adolescents in South Africa have larger gaps at each step of the HIV continuum of care resulting in low levels of viral suppression.

**Methods:** We conducted a systematic review and modified Delphi analysis of interventions addressing the HIV continuum of care for adolescents in South Africa. We searched PubMed, Science Direct, and Google Scholar and online conference proceedings from the International AIDS Society, the International AIDS Conference, and the Conference on Retrovirology and Opportunistic Infections from January 1, 2010 to September 30, 2020. We then conducted a modified Delphi analysis with 29 researchers involved in the National Institutes of Health's Fogarty International supported Adolescent HIV Implementation Science Alliance – South Africa to evaluate interventions for efficacy, feasibility, and potential for scale-up.

**Results:** We identified nine initial published articles containing interventions addressing the adolescent HIV continuum of care in South Africa, including five interventions focused on HIV diagnosis, two on antiretroviral therapy adherence, and two on retention in care. No studies

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addressed linkage to care or transition from pediatric to adult care. Two studies discussed intervention costs. In-home and HIV-self testing, community-based adherence support, and provision of adolescent-friendly services were the most impactful and scalable interventions addressing the adolescent HIV continuum of care.

**Conclusion:** Future interventions should work comprehensively across the adolescent HIV continuum of care and be tailored to the specific needs of adolescents.

## Strengths and Limitations of the Study

- The AHI(SA)$^2$ network includes highly experienced experts with key knowledge to evaluate interventions for adolescents and young adults living with HIV in South Africa.
- We evaluated interventions with quantifiable results addressing the full continuum of care for adolescents and young adults living with HIV in South Africa.
- The evaluation included interventions that were published in the literature and did not include studies that were in progress or planned.
- Not all AHI(SA)$^2$ members who were invited to participate completed the survey.

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**Introduction:**

South Africa has the highest burden of adolescents and young adults living with HIV (ALWH) in the world (1). In the late 1990s and early 2000s, delays in access to HIV treatment and suboptimal HIV prevention tools contributed to more than 330,000 children being born with HIV through perinatal transmission (2-5). These children are now aging into adolescence and young adulthood. In addition, non-perinatal (typically sexual) transmission is a major concern among adolescents in South Africa. In 2017 alone, an estimated 50,200 adolescents aged 15-19 contracted HIV, with the number of females (n=41,000) being more than four times higher than males (n=9,200) (1). Despite growing recognition of the challenges in adolescent HIV prevention and care, HIV/AIDS remains the leading cause of death among all adolescents in South Africa (6).

South Africa has the largest antiretroviral therapy (ART) program in the world, and national ART guidelines support early initiation of ART with simple, potent, and well-tolerated regimens (7). ART scale-up has contributed to relatively high levels of viral suppression, low mortality rates, and low HIV transmission among adults living with HIV in South Africa (1, 8, 9). However, the continuum of care for ALWH (i.e., from diagnosis to linkage to care and subsequent ART adherence, retention in care and viral suppression) has significant gaps at each step, resulting in only 10% of adolescents who are living with HIV in South Africa being virally suppressed in a study conducted in 2016 (2). In addition, the process of transition from pediatric to adult care lacks evidence-based guidelines and has shown poor levels of retention in care and viral suppression after transition (10-12).

While research is ongoing to support HIV treatment for ALWH in South Africa, the extent and efficacy of interventions has not been well defined, and evidence-based guidance is needed to direct future intervention development. Our objective was to utilize a network of researchers working to improve HIV care for ALWH in South Africa to identify promising interventions for wider

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implementation and future scale up of successful interventions for this population. Here, we present the results of a systematic review and modified Delphi analysis of interventions addressing the adolescent and young adult HIV continuum of care in South Africa.

## Methods

*Systematic review*

We searched PubMed, Science Direct, and Google Scholar and online conference proceedings from the International AIDS Society (IAS), the International AIDS Conference (AIDS), and the Conference on Retrovirology and Opportunistic Infections (CROI) from January 1, 2010 to September 30, 2020. Key words and medical subject headings relevant to age (i.e., adolescent, adolescence, teen, young adult) were cross-referenced with terms associated with the HIV continuum of care (i.e., diagnosis, linkage to care, retention, adherence, and transition). Interventions addressing HIV prevention among adolescents and young adults were analyzed separately and will be published separately. We reviewed literature published in English that reported interventions addressing the adolescent and young adult HIV continuum of care that were piloted or implemented in South Africa among adolescents and young adults. We adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (supplement 1) (13). Because of variability in reported age ranges, we included studies in our analysis with primary data on youth ages 10 to 24 years. We present age ranges as described by the original authors.

All identified studies were uploaded into Covidence, a non-profit website working with the Cochrane database to improve systematic reviews (www.covidence.org, Melbourne, Australia). Duplicate materials were removed. After initially screening the titles and abstracts, two authors (BCZ and TR) independently reviewed potential studies. Conflicts were resolved by reviewing the full text article and discussing the inclusion/exclusion criteria until consensus was reached. We



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included randomized controlled trials, pre-post evaluations, and retrospective cohorts that evaluated interventions with primary outcomes addressing the adolescent HIV continuum of care in South Africa. We excluded review articles, study protocols, qualitative studies and studies that did not include an intervention, did not report primary data affecting the HIV continuum of care, or were targeting children or adults outside our specified age range of 10 to 24 years. We then extracted data including study population, location, design, intervention description, and outcomes from the full text articles and created summarized tables organized by each step in the HIV continuum of care.

We used the PRISMA guidelines to assess the quality of the study, strength of evidence and bias for all included clinical trials and evaluated random sequence generation, allocation concealment, blinding, and outcome reporting (13). Randomized controlled trials were considered less biased than pre-post evaluations, and prospective and retrospective observational studies. Observational studies were evaluated for bias using the GRADE guidelines and each article included an assessment of eligibility, controls, loss to follow-up, and outcome consistency (14).

*Modified Delphi analysis*

To identify the most effective, feasible and scalable interventions impacting the adolescent HIV continuum of care in South Africa, we engaged the National Institute of Health's Fogarty International supported Adolescent HIV Implementation Science Alliance – South Africa (AHI(SA)$^2$ (www.ahisa2.org). Members of AHI(SA)$^2$ were selected by inviting (via email) researchers with current NIH funding and/or published first authors (and their research teams) addressing adolescent HIV prevention or the adolescent HIV continuum of care in South Africa. We uploaded summarized tables from our systematic review, along with links to the full text articles and abstracts, into REDCap for participants to access (15). We invited AHI(SA)$^2$ members to complete a questionnaire assessing the feasibility, efficacy, scalability, and costing of each intervention (using four-point Likert scales). We also asked members to assign an overall score



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(rating of 0 to 100) regarding the potential impact of the intervention on the continuum of care for ALWH in South Africa. AHI(SA)$^2$ leadership decided *a priori* that a score of 70 or greater would be a reasonable estimate of clinically meaningful effects and would be required to be included in the second-round analysis.

All AHI(SA)$^2$ members (n=42) received a secure link to the REDCap survey via email to complete the first-round questionnaire focused on intervention efficacy (August 24, 2020 – September 23, 2020). Results were discussed at the first AHI(SA)$^2$ virtual meeting on September 25, 2020 with the 25 participants who attended. Interventions with a prespecified median overall impact score less than 70 or with an average efficacy score less than three (out of four on the Likert scale) were removed from the next round questionnaire. In preparation for the second round, we updated the literature search from July 31, 2020 to September 30, 2020 and invited AHI(SA)$^2$ members to nominate additional studies for inclusion. The REDCap questionnaire was updated and emailed to AHI(SA)$^2$ members who participated in the first-round questionnaire (n=29) to evaluate feasibility and scalability (September 25, 2020 to December 15, 2020). We sought to formulate an expert consensus on the interventions for ALWH with highest potential be taken to scale and to highlight areas in which future interventions are needed (16, 17).

*Patient and public involvement*

Patients or the public were not involved in the design or conduct of this study. The research questions were formulated by the AHI(SA)$^2$ leadership team. Results of this analysis were presented at regular AHI(SA)$^2$ meetings.

## Results

*Systematic review*

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Our initial search yielded 273 relevant articles. Of these, 110 were removed due to duplicate publications or interventions, and 112 did not include interventions addressing the adolescent HIV continuum of care in South Africa, leaving 25 full text articles for review. We excluded 16 of the 25 studies because nine lacked an intervention, three focused on HIV primary prevention, and four met one of the following exclusionary criteria: literature review only (n=1), no primary results (n=1), not focused on adolescents (n=1), or not conducted in South Africa (n=1). We did not find any additional unpublished studies in the review of conference abstracts. Overall, nine published papers with sufficient data were included in the first round of the modified Delphi analysis addressing the adolescent and young adult HIV continuum of care in South Africa.

*Modified Delphi analysis*

Participants: AHI(SA)[2] consisted of 42 eligible individuals involved in adolescent HIV research in South Africa. For the first round of the modified Delphi analysis, 29 (69%) participated in the REDCap survey. Participant characteristics are presented in **Table 1**. Of the 29 who initiated the survey, 21 (72%) completed it. Of the 29 participants who were sent the second-round survey, 21 (72%) initiated it and 18 (62%) completed the survey.

**Table 1: Participant characteristics in two rounds of the modified Delphi Analysis.**

Publications:

Among the nine initial published articles containing interventions addressing the adolescent HIV continuum of care in South Africa, five interventions focused on HIV diagnosis, two on ART adherence, and two on retention in care as indicated in **Supplementary Table 1.** No studies addressed linkage to care or transition from pediatric to adult care. For study design, five studies were randomized controlled trials, three were retrospective cohort analyses, and one was a pre-post evaluation. Two studies of the nine discussed the cost of the intervention.

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Intervention evaluation:

Below we report summaries of the individual published interventions for ALWH in South Africa separated by steps in the continuum of care. Within each step, the interventions are listed beginning with the highest overall median impact rating in the modified Delphi Analysis (**Supplementary Table 1 and Table 2**).

**Supplementary Table 1:** Interventions to improve the continuum of care for adolescents and young adults living with HIV in South Africa

**Table 2:** Second round modified Delphi analysis of interventions to improve the continuum of care for adolescents living with HIV in South Africa

HIV Diagnosis (n=5)

Five interventions addressed HIV diagnosis through expanded HIV testing among adolescents. Four of these were randomized controlled trials (18-21) and one was a pre-post evaluation (22). Three of these studies met criteria for inclusion in both rounds of the modified Delphi analysis(18, 19, 21) Two of the three studies showed high rates of HIV testing uptake and acceptability of in-home HIV testing and one showed strong short-term effects.

Shanaube et al. reported on the 11,175 adolescents randomized to receive the full intervention of door-to-door HIV testing and counseling with support for linkage to care delivered by community health workers compared to standard care in the Zambia PopART for Youth study (21). HIV testing was done for 80% of participants with awareness of a positive HIV test increased by a three-fold by the third round of interventions and a median time to ART linkage of five months; however, there was no reported comparison to standard of care for HIV testing. Doherty et al. performed a cluster randomized clinic trial among adolescents evaluating door-to-door home-based HIV testing performed by lay counselors compared to standard of care (18). HIV testing was significantly higher among in adolescents receiving home-based testing (69%) compared to standard of care



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(47%) (prevalence ratio 1.54, 95% CI 1.32 to 1.81). Pettifor et al. showed high uptake of HIV self-testing among young women. This study randomized 287 young women ages 18 to 26 to choose between HIV self-testing kits or receive clinic-based HIV testing compared to HIV testing in government clinics (19). In the choice arm, 95% chose self-testing. After three months, 92% of the choice arm and 43% of the standard of care arm reported HIV testing a difference of 49%. By nine months, the difference reduced to 25%.

The remaining two studies did not meet the threshold to continue to the second round of the modified Delphi analysis due to lack of demonstrated efficacy as determined by the first round of the modified Delphi analysis; both used soccer as a means to increase HIV testing. Hershow et al. conducted a pre-post analysis of the SKILLZ Street intervention by Grassroot Soccer where female coaches delivered afterschool education program consisting of 10 two-hour bi-weekly sessions with 1,953 female adolescents ages 12 to 16 years (22). Among the participants, 69% tested for HIV by the end of the intervention. There was no control group or comparison to baseline testing. Rotheram-Borus et al. randomized 142 male adolescents between the ages of 18 and 25 to immediately participate in Grassroot Soccer with trained coaches, rapid HIV testing, and vocational training compared to delayed participation in the control group (20). There was no significant difference in HIV testing among the immediate group (29%) compared to the delayed uptake group (24%) after the intervention.

ART adherence (n = 2)

Two studies evaluated interventions designed to improve adolescent ART adherence (23, 24). Both were included in first round of the modified Delphi analysis, but only one continued to the second round due to lack of perceived efficacy. Cluver et al. evaluated the provision of social protections on adherence among 1,059 adolescents ages 10 to 19 years in an observational cohort (24). They found that food provision (adjusted odds ratio [aOR] 0.57, 95% CI 0.42 – 0.76), attendance at HIV



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support groups (aOR 0.60, 95% CI 0.40 – 0.91) and parental supervision (aOR 0.46, 95% CI 0.43 – 0.73) were associated with lower rates of self-reported non -adherence to ART in the prior week. Bhana et al.'s VUKA Family Program was not included in the second round of the modified Delphi analysis due to lack of clear efficacy. The intervention consisted of 10 sessions over three months and used a culturally tailored cartoon storyline of step-by-step guidance for problem solving skills and facilitated family discussions (23). ART adherence (last time missed medication) increased over time from baseline to completion of the study, but there was no comparison to standard of care.

Retention in care (n=2)

Two studies were designed to improve retention in care, and both were included in the first and second rounds of the modified Delphi Analysis. Zanoni et al. evaluated a weekend-based, adolescent-focused clinic that offered peer support, group activities, counseling, and meals compared to standard of care in a retrospective cohort analysis (25). Retention in care and viral suppression were higher among adolescents who attended the adolescent-friendly clinic (95% and 91%, respectively) compared to standard of care (85% and 80%, respectively; aOR 3.5, 95% CI 1.2 – 11.1; p=0.018 and aOR 2.5, 95% CI 1.1 – 5.8; p=0.028, respectively). The second study by Fatti et al. evaluated community-based support that included home-based ART education, psychosocial support, opportunistic infection screening, and access to government grants compared to standard of care among 6,706 adolescents aged 10 to 24 years (26). Cumulative loss to follow up was lower in the intervention group (30%) compared to standard of care (39%; aHR = 0.60; 95% CI 0.51 – 0.71; p <0.0001). Although there was no statistically significant difference in viral failure after three or five years; 8.8% in the intervention group had viral failure compared to 37.2% in the standard of care group (aOR 0.24, 95%CI 0.06 – 1.03), the difference may be clinically meaningful.

**Discussion:**

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In our systematic review and modified Delphi analysis of interventions addressing the adolescent HIV continuum of care, we found five feasible and effective interventions with high potential for scale-up to improve adolescent HIV outcomes in South Africa. To improve HIV diagnosis among ALWH in South Africa, in-home HIV testing (19) and HIV self-testing (21) were promising interventions. However, there was a lack of demonstrated efficacy in targeting successful linkage to care after HIV testing. Adherence to ART among adolescents improved with the provision of food support, attendance in HIV support groups, and parental supervision (24). Two interventions improved retention in care: one focused on adolescent-friendly services by providing an after-school hours clinic with peer support, counseling, and meals (25) and the other by providing in-home support for HIV education, psychosocial support, and assistance with applications to governmental grants (26). Unfortunately, no interventions addressed transition from pediatric to adult based care for ALWH in South Africa.

Although adolescents and young adults have the highest incidence of new HIV infections in South Africa, they often do not present to health centers for HIV testing. Structural barriers such as school, transportation, or finances; psychosocial barriers such as the feelings of invincibility; or clinical factors such as the lack of clinical symptoms all contribute to poor engagement in care (1, 6). Therefore, traditional methods including clinic-based opt-out testing are not ideal for this population. Non-traditional methods of intensified case finding are required to identify adolescents living with undiagnosed HIV. In-home HIV testing provided by lay counselors offers a solution to the issues of adolescents not attending health centers. Although this process can be labor intensive and costly, it also offers a mechanism to improve linkage to care once adolescents are diagnosed with HIV. Another potential solution to low HIV testing rates among adolescents and young adults is self-testing which has shown acceptability and uptake but requires interventions to provide counseling to newly diagnosed individuals and facilitate linkage to care.

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Adolescents living with HIV commonly suffer poor adherence to ART (27, 28). Direct interventions addressing adherence among adolescents living with HIV are often require considerable effort and resources and have variable efficacy (29, 30). Indirect measures that improve food and economic security, psychosocial support, and supervision appear to improve adherence among adolescents (24). Future interventions to address adherence should include economic and psychosocial support. In addition, addressing barriers such as depression and substance use which has improved ART adherence in adults may also benefit adolescents and young adults (31).

After-school hours clinics, peer support, and improved connection to clinical staff have been shown to improve retention in care among adolescents living with HIV (32). As recommended by the World Health Organization, the provision of adolescent-friendly services and reimagining of clinical care delivery for adolescents and young adults can strengthen retention (33). For example, many of these services can be provided outside of clinic space and time. In-home or virtual clinical services could be expanded to improve adolescent retention in care.

Interventions require resources in time, personnel, and funding but only two interventions in this review presented cost analyses. For implementors and policymakers to adequately assess intervention scale-up, costing and cost-effectiveness analyses are required. Future interventions addressing the adolescent HIV continuum of care should evaluate cost.

We identified several interventions that could be scaled to improve the adolescent HIV continuum of care in South Africa; however, several gaps remain. Improving HIV diagnosis is critical to improve the continuum of care, but without linkage to care, adolescents cannot access ART and achieve viral suppression. Interventions to improve adolescent HIV testing should include linkage to care, and adolescents require additional support to achieve and maintain viral suppression. Moreover, children initiating ART under the care of pediatricians will eventually require transition

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to adult care; however, we did not find any published interventions addressing transition care for ALWH in South Africa highlighting a key area for future research.

Future interventions addressing the adolescent HIV continuum of care should build on the promising results from the highlighted studies and perhaps include a combination of the most effective strategies to optimize outcomes. Additionally, acceptability of the interventions by adolescents is critical for continued success of potential interventions. (34) Using participant-centered design and direct input from adolescent early in intervention development can likely increase acceptability and improve likelihood of intervention success (35).

This systematic review and modified Delphi analysis has several limitations. First, we only identified interventions that were published in the literature and did not include studies that were in progress or planned. In addition, the modified Delphi analysis did not include researchers who were not yet published, received non-NIH funding, or did not respond to email invitations. However, the AHI(SA)$^2$ network includes highly experienced experts with key knowledge to evaluate interventions in South Africa. In addition, not all AHI(SA)$^2$ members who were invited to participate completed the survey. We experienced drop-out rates of 28% and 38% in rounds one and two of the modified Delphi analysis due to strict completion deadlines prior to scheduled AHI(SA)$^2$ meetings. However, this review highlights published interventions with quantifiable results addressing the continuum of care for adolescents and young adults evaluated by scholars with interest in this population.

**Conclusion:**

No single intervention exists to improve each step of the continuum of care for ALWH in South Africa. However, we highlight several promising interventions, specifically in-home HIV testing and self-testing to increase the proportion of ALWH who know their diagnosis and economic



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support and food support to increase ART adherence. These interventions should be further evaluated as they are implemented to improve adolescent outcomes in the HIV care cascade. Future interventions should work comprehensively across the adolescent HIV continuum of care and be tailored to the specific needs of adolescents.

**Table 1: Participant characteristics in two rounds of the modified Delphi Analysis.**

| Demographics | First Round (N = 29) | Second Round (N=21) |
|---|---|---|
| | n (%) | n (%) |
| **Profession** | | |
| Researcher | 13 (45%) | 12 (57%) |
| Clinician/researcher | 6 (21%) | 4 (19%) |
| Medical doctor | 6 (21%) | 3 (14%) |
| Healthcare provider | 2 (7%) | 1 (5%) |
| Student/fellow | 2 (7%) | 1 (5%) |
| **Years working in adolescent HIV** | | |
| > 10 years | 14 (48%) | 10 (48%) |
| 6 – 10 years | 6 (21%) | 6 (29%) |
| 1 – 5years | 8 (28%) | 4 (19%) |
| < 1 year | 1 (3%) | 1 (5%) |
| **Highest Educational Degree** | | |
| PhD | 11 (38%) | 9 (43%) |
| MD or MBChB | 5 (17%) | 2 (10%) |
| MPH or other master's degree | 5 (17%) | 4 (19%) |
| MD/MBChB + PhD | 3 (10%) | 3 (14%) |
| MD/MBChB + master's degree | 2 (7%) | 3 (14%) |
| Other | 3 (10%) | --- |
| **Primary residence** | | |
| South Africa | 22 (76%) | 15 (71%) |
| United States | 6 (21%) | 4 (19%) |
| United Kingdom | 1 (3%) | 1 (5%) |

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**Table 2:** Second round Delphi analysis of interventions to improve the continuum of care for adolescents living with HIV in South Africa

| Authors | Title | Publication and date | Step in continuum of care | Costing | Feasibility mean score | Efficacy mean score | Scalability mean score | Overall median impact score |
|---|---|---|---|---|---|---|---|---|
| Zanoni, et al. | Higher Retention and Viral Suppression with Adolescent-Focused HIV Clinic in South Africa | *PLoS One* 2017 | Retention | $1.25 per adolescent per clinic visit | 3.34 | 3.28 | 3.17 | 80 |
| Fatti, et al. | The Effectiveness and Cost-Effectiveness of Community-Based Support for Adolescents Receiving Antiretroviral Treatment: An Operational Research Study in South Africa | *Journal of the International AIDS Society* 2018 | Retention | Cost of intervention was US$49.50/ patient/year. Incremental cost per patient-loss averted was US$600 and US$776 after 1 and 2 years, respectively. | 3.00 | 2.83 | 3.11 | 76 |
| Shanaube, et al. | Community Intervention Improves Knowledge of HIV Status of Adolescents in Zambia: Findings from HPTN 071-PopART for Youth Study | *AIDS* 2017 | HIV Diagnosis | Not evaluated | 3.39 | 3.11 | 3.00 | 75 |
| Pettifor, et al | HIV Self-testing among Young Women in Rural South Africa: A randomized controlled trial comparing clinic-based HIV testing to the choice of either clinic testing or HIV self-testing with secondary distribution to peers and partners | *EClinical Medicine* 2020 | HIV Diagnosis | Not evaluated | 3.06 | 2.76 | 3.00 | 72 |
| Cluver, et al. | Achieving Equity in HIV-Treatment Outcomes: Can Social Protection Improve Adolescent ART-Adherence in South Africa? | *AIDS Care* 2016 | ART Adherence | Not evaluated | 3.00 | 2.94 | 2.67 | 68 |
| Doherty, et al. | Effect of Home-Based HIV Counseling and Testing Intervention in Rural South Africa: Cluster Randomized Trial | *BMJ* 2013 | HIV Diagnosis | Not evaluated | 3.17 | 3.06 | 2.94 | 65 |

**Declarations:**

**Author's Contributions:**

Dr. Zanoni conceptualized and designed the study, performed the literature review, conducted the analysis, drafted the initial manuscript, reviewed and revised the manuscript and approved the final manuscript as submitted.

Ms. Ramos performed the literature review, assisted with the analysis, assisted in the drafting of the initial manuscript, reviewed and revised the manuscript and approved the final manuscript as submitted.



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Dr. Archary assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Ms. Sibaya assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Donenberg assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Shahmanesh assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Celum assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Pettifor assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Bekker assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Haberer assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

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**Availability of data:** The datasets used during the current study are available from the corresponding author on reasonable request.

**Ethics:** The protocol was evaluated by the Emory University Institutional Research Ethics Board and given a waiver of formal ethics review.

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**SupplementaryTable 1:** Interventions to improve the continuum of care for adolescents and young adults living with HIV in South Africa

| Authors | Title | Publication and Date | Location | Number of Subjects | Ages (Target Population) | Study Type | Intervention Description | Overall Results | Median Overall Score Round 1 |
|---|---|---|---|---|---|---|---|---|---|
| **Adolescent HIV Testing** | | | | | | | | | |
| Doherty, et al. | Effect of Home-Based HIV Counseling and Testing Intervention in Rural South Africa: Cluster Randomised Trial | *BMJ* 2013 | uMzimkhulu subdistrict, KwaZulu-Natal, South Africa | 4154 Adolescents and Adults | 14-17, 18+ Adult Guardians | Cluster Randomized Controlled Trial; Door-to-Door Testing | Door-to-door home-based HIV testing (HBT) performed by lay counselors compared to standard of care HIV counseling and testing at local clinics. | Testing completed in 69% of HBT compared to 47% in control (PR 1.54 95% CI 1.32 to 1.81) | 72.5 |
| Rotheram-Borus, et al. | Feasibility of Using Soccer and Job Training to Prevent Drug Abuse and HIV | *AIDS and Behavior* 2016 | South Africa | 142 Adolescents | 18-25 Unemployed Males | Community Randomized Controlled Trial | Randomized neighborhoods to immediate intervention or delayed control; Intervention: Grassroot Soccer program with trained coaches, random rapid diagnostic tests for alcohol/drugs, and vocational training. | 29% testing in immediate vs. 24% in delayed (no significant difference) | 50 |
| Hershow, et al. | Using Soccer to Build Confidence and Increase HCT Uptake Among Adolescent Girls | *Sport in Society* 2015 | South Africa | 1953 Adolescents | 12-16 (Females Only) | Pre-Post Evaluation | Survey before and after participation in SKILLZ Street intervention developed by Grassroot Soccer (GRS). SKILLZ Street female coaches delivered an afterschool education program consisting of 10, two hour biweekly sessions. | 69% tested for HIV<br>No control group. | 63 |
| Shanaube, et al. | Community Intervention Improves Knowledge of HIV Status of Adolescents in Zambia: Findings from HPTN 071-PopART for Youth Study | *AIDS* 2017 | Results from Zambia only; Intervention also conducted in-South Africa | 11 175 Adolescents | 15-19 | Community-Randomized Clincal Trial | **HPTN 071 (PopART)**<br>Door-to-door HIV prevention package delivered by community health worders. Intervention included in-home HIV Counseling and Testing (HCT), support for linkage to care for individuals identified living with HIV, ART adherence support, referrals for voluntary medical male circumcision | HCT uptake 80% (8,707/10,809) and awareness of an HIV-positive status tripled from 75 to 210. Home visits from providers saw the percentage of ALWH aware of their HIV status rise from 27.6% to 88.5%.<br><br>No comparison with standard care. | 76 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Pettifor, et al | HIVself-testing among young women in rural South Africa: A randomized controlled trial comparing clinic-based HIVtesting to the choice of eitherclinict esting or HIVself-testing with secondary distribution to peers andartners | *EClinical Medicine* 2020 | Mpumlanga, SouthAfrica | 287 women | 18 - 26 | Randomized Cotrolled Trial | Standard of care HIV counseling and testing (HCT) at government clinics vs. choice of clinic based HCT or HIV self-testing kits.<br><br>+referrals for peers to recelive clinic based HCT or HIV self-testing kits | 146 randomized to HCT and 141 to choice. In the choice arm 95% (n=135) chose HIV self testing kits. After 3 months a significantly higher percent (92%) of participants in the choice arm reported HIV testing compared to (43%) the HCT arm with a difference of 43% (95% CI 40% - 58%). By 9 months the difference decreased to 25% (95% CI 17% - 33%). Participants in the choice arm were more likely to invite peers or partners to test (94% vs. 76%) | 70 |
| **Adherence** | | | | | | | | | |
| Bhana, et al. | The VUKA Family Program: Piloting a Family-Based Psychosocial Intervention to Promote Health and Mental Health Among HIV Infected Early Adolescents in South Africa | *AIDS Care* Jan 2014 | KwaZulu-Natal, South Africa | 65 Adolescents and Their Family Members | 10-13 | Randomized Controlled Trial | **VUKA Family Program** Inspired by CHAMP, the intervention consisted of 3 months and ten sessions involving a culturally-tailored cartoon storyline and curriculum. The curriculum follows the story and provdes step-by-step guidance for problem solving and facilitating discussions. ALWH were assessed for feelings of stigma, knowledge of ART, adherence depression and caregiver support. | ART adherence: Intervention (I) 1.10, Control (C) -. 43 at baseline; Follow-up, $p < .05$.<br><br>Caregiver/child communication: (I) .41, (C) -.07, $p < .09$.<br><br>Youth knowledge of HIV treatment: (I) .27; (C) -.74, $p < .08$. | 68.5 |
| Cluver, et al. | Achieving Equity in HIV-Treatment Outcomes: Can Social Protection Improve Adolescent ART-Adherence in South Africa? | *AIDS Care* 2016 | Eastern Cape, South Africa | 1059 Adolescents | 10-19 | Observational Cohort | Evaluated the provision of social protections on adherence. Social provisions included government cash transfers, food security, school fees/materials, school feeding, and clothing; HIV support group, sports groups, choir/art groups, positive parenting and parental supervision/monitoring. | Three social protection provisions were associated with reduced non-adherence:<br><br>Food provision (aOR .57, CI .42–.76, p < .001);<br><br>HIV support group attendance (aOR .60, CI .40–.91, p < .02); High parental/caregiver supervision (aOR .56, CI .43–.73, p < .001). | 75 |
| **Retention in Care** | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|
| Fatti, et al. | The Effectiveness and Cost-Effectiveness of Community-Based Support for Adolescents Receiving Antiretroviral Treatment: An Operational Research Study in South Africa | *Journal of the International AIDS Society* 2018 | KwaZulu-Natal/Western Cape/Eastern Cape/Mpumalanga, South Africa | 6706 Adolescents | 10-24 | Retrospective Cohort Study | Community based support (CBS) included home-based ART-related education, psychosocial support, symptom screening for opportunistic infections and support to access government grants. CBS was compared to standard of care. | Cumulative LTFU was 40% lower amongst participants receiving CBS (29.9%) compared to participants without CBS (38.9%), aHR = 0.60 (95% CI: 0.51 to 0.71); p < 0.0001).<br><br>Viral failure: at 3 years – no difference; at five years 18.8% CBS participants versus 37.2% non-CBS participants had viral failure, adjusted odds ratio = 0.24 (95% CI: 0.06 to 1.03). | 80 |
| Zanoni, et al. | Higher Retention and Viral Suppression with Adolescent-Focused HIV Clinic in South Africa | *PLoS One* 2017 | KwaZulu-Natal, South Africa | 254 HIV+ Adolescents | 13-24 | Retrospective Cohort Analysis | Weekend adolescent-focused clinic with peer support, group activities, counseling, and meals compared to standard of care. | Retention: 95% in adolescent-friendly clinic vs. 85% in standard clinic (OR 3.5; 95% CI 1.2 – 11.1; p=0.018).<br><br>Viral Suppression: 91% adolescent-friendly clinic vs. 80% in standard clinic. (OR 2.5; 95% CI 1.1 – 5.8; p=0.028 | 88 |

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Appendix 1: Search terms:

Pubmed search strategy:

((adolescent[MeSH Terms]) OR (teen*) OR (youth*) OR (adoles*) AND ((HIV infection[MeSH Terms]) OR (acquired immune deficiency syndrome[MeSH Terms]) OR (HIV[TIAB]) OR ("AIDS"[TIAB]) OR ("acquired immune deficiency syndrome"[TIAB]) OR ("acquired immunodeficiency syndrome"[TIAB])) AND ("south africa"[TIAB]) AND ((intervention[TIAB])

Google scholar search strategy:
"adolescen* HIV intervention South Africa"

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# Zanoni BMC Open | PRISMA Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a literature review. | Title page 1 |
| **ABSTRACT** | | | |
| Abstract | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings.<br>See the PRISMA 2020 for Abstracts checklist for the complete list. | Abstract page 1 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge, i.e., what is already known about your topic. | Page 2, lines 23 - 29 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | Pages 2 – 3, lines 25 - 29 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses with study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | Page 3, lines 33 - 46 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | Page 3, lines 33 - 39 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | Appendix 1 |
| Selection process | 8 | State the process for selecting studies (i.e., screening, eligibility).<br>Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | Pages 3 – 4, lines 48 - 59 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | Page 4, lines 61 - 66 |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | Pages 5 – 6, lines 99 - 107 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were | N/A |

The checklist has been adapted for KIN 4400 Independent Research Study in Kinesiology at the University of Guelph-Humber. Last updated: Dec 9, 2021

*Adapted From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71 *Adapted From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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## Zanoni BMC Open | PRISMA Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| | | excluded. | |
| Study characteristics | 17 | Cite each included study and present its characteristics (e.g., study size, PICOS, follow-up period). | Table 1 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | Page 4, lines 61 - 66 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Table 1 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | Pages 9 - 12 |
| | 23b | Discuss any limitations of the evidence included in the review. | Page 12, lines 267 - 277 |
| | 23c | Discuss any limitations of the review processes used. | Page 12, lines 267 - 277 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | Page 12, lins 280 - 286 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | N/A |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | Not prepared |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | None |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | Supplementary information |
| Competing interests | 26 | Declare any competing interests of review authors. | Supplementary information |
| Availability of data, code, and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | Supplementary information |

The checklist has been adapted for KIN 4400 Independent Research Study in Kinesiology at the University of Guelph-Humber. Last updated: Dec 9, 2021

*Adapted From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71 *Adapted From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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# Interventions addressing the adolescent HIV continuum of care in South Africa: a systematic review and modified Delphi analysis

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# Interventions addressing the adolescent HIV continuum of care in South Africa: a systematic review and modified Delphi analysis

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**Abstract:**

**Introduction:** Compared to adults, adolescents in South Africa have larger gaps at each step of the HIV continuum of care resulting in low levels of viral suppression.

**Methods:** We conducted a systematic review and modified Delphi analysis of interventions addressing the HIV continuum of care for adolescents in South Africa. We searched PubMed, Science Direct, and Google Scholar and online conference proceedings from the International AIDS Society, the International AIDS Conference, and the Conference on Retrovirology and Opportunistic Infections from January 1, 2010 to September 30, 2020. We then conducted a modified Delphi analysis with 29 researchers involved in the National Institutes of Health's Fogarty International supported Adolescent HIV Implementation Science Alliance – South Africa to evaluate interventions for efficacy, feasibility, and potential for scale-up.

**Results:** We identified nine initial published articles containing interventions addressing the adolescent HIV continuum of care in South Africa, including five interventions focused on HIV diagnosis, two on antiretroviral therapy adherence, and two on retention in care. No studies addressed linkage to care or transition from pediatric to adult care. Two studies discussed intervention costs. In-home and HIV-self testing, community-based adherence support, and provision of adolescent-friendly services were the most impactful and scalable interventions addressing the adolescent HIV continuum of care.

**Conclusion:** Future interventions should work comprehensively across the adolescent HIV continuum of care and be tailored to the specific needs of adolescents.

## Strengths and Limitations of the Study

- The AHI(SA)$^2$ network includes highly experienced experts with key knowledge to evaluate interventions for adolescents and young adults living with HIV in South Africa.
- We evaluated interventions with quantifiable results addressing the full continuum of care for adolescents and young adults living with HIV in South Africa.
- The evaluation included interventions that were published in the literature and did not include studies that were in progress or planned.
- Not all AHI(SA)$^2$ members who were invited to participate completed the survey.

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**Introduction:**

South Africa has the highest burden of adolescents and young adults living with HIV (ALWH) in the world (1). In the late 1990s and early 2000s, delays in access to HIV treatment and suboptimal HIV prevention tools contributed to more than 330,000 children being born with HIV through perinatal transmission (2-5). These children are now aging into adolescence and young adulthood. In addition, non-perinatal (typically sexual) transmission is a major concern among adolescents in South Africa. In 2017 alone, an estimated 50,200 adolescents aged 15-19 contracted HIV, with the number of females (n=41,000) being more than four times higher than males (n=9,200) (1). Despite growing recognition of the challenges in adolescent HIV prevention and care, HIV/AIDS remains the leading cause of death among all adolescents in South Africa (6).

South Africa has the largest antiretroviral therapy (ART) program in the world, and national ART guidelines support early initiation of ART with simple, potent, and well-tolerated regimens (7). ART scale-up has contributed to relatively high levels of viral suppression, low mortality rates, and low HIV transmission among adults living with HIV in South Africa (1, 8, 9). However, the continuum of care for ALWH (i.e., from diagnosis to linkage to care and subsequent ART adherence, retention in care and viral suppression) has significant gaps at each step, resulting in only 10% of adolescents who are living with HIV in South Africa being virally suppressed in a study conducted in 2016 (2). In addition, the process of transition from pediatric to adult care lacks evidence-based guidelines and has shown poor levels of retention in care and viral suppression after transition (10-12).

While research is ongoing to support HIV treatment for ALWH in South Africa, the extent and efficacy of interventions has not been well defined, and evidence-based guidance is needed to direct future intervention development. Our objective was to utilize a network of researchers working to improve HIV care for ALWH in South Africa to identify promising interventions for wider



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implementation and future scale up of successful interventions for this population. Here, we present the results of a systematic review and modified Delphi analysis of interventions addressing the adolescent and young adult HIV continuum of care in South Africa.

## Methods

*Systematic review*

We searched PubMed, Science Direct, and Google Scholar and online conference proceedings from the International AIDS Society (IAS), the International AIDS Conference (AIDS), and the Conference on Retrovirology and Opportunistic Infections (CROI) from January 1, 2010 to September 30, 2020. Key words and medical subject headings relevant to age (i.e., adolescent, adolescence, teen, young adult) were cross-referenced with terms associated with the HIV continuum of care (i.e., diagnosis, linkage to care, retention, adherence, and transition) (**Supplement 1**). Interventions addressing HIV prevention among adolescents and young adults were analyzed separately and will be published separately. We reviewed literature published in English that reported interventions addressing the adolescent and young adult HIV continuum of care that were piloted or implemented in South Africa among adolescents and young adults. We adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (**Supplement 2**) (13). Because of variability in reported age ranges, we included studies in our analysis with primary data on youth ages 10 to 24 years. We present age ranges as described by the original authors.

All identified studies were uploaded into Covidence, a non-profit website working with the Cochrane database to improve systematic reviews (www.covidence.org, Melbourne, Australia). Duplicate materials were removed. After initially screening the titles and abstracts, two authors (BCZ and TR) independently reviewed potential studies. Conflicts were resolved by reviewing the full text article and discussing the inclusion/exclusion criteria until consensus was reached. We



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included randomized controlled trials, pre-post evaluations, and retrospective cohorts that evaluated interventions with primary outcomes addressing the adolescent HIV continuum of care in South Africa. We excluded review articles, study protocols, qualitative studies and studies that did not include an intervention, did not report primary data affecting the HIV continuum of care, or were targeting children or adults outside our specified age range of 10 to 24 years. We then extracted data including study population, location, design, intervention description, and outcomes from the full text articles and created summarized tables organized by each step in the HIV continuum of care.

We used the GRADE guidelines to assess the quality of the study, strength of evidence and bias for all included clinical trials and evaluated random sequence generation, allocation concealment, blinding, and outcome reporting (14). Randomized controlled trials were considered less biased than pre-post evaluations, and prospective and retrospective observational studies. Observational studies were evaluated for bias using the GRADE guidelines and each article included an assessment of eligibility, controls, loss to follow-up, and outcome consistency (15). (See **Supplement 3** for quality assessment table)

*Modified Delphi analysis*

To identify the most effective, feasible and scalable interventions impacting the adolescent HIV continuum of care in South Africa, we engaged the National Institute of Health's Fogarty International supported Adolescent HIV Implementation Science Alliance – South Africa (AHI(SA)$^2$ (www.ahisa2.org). Members of AHI(SA)$^2$ were selected by inviting (via email) researchers with current NIH funding and/or published first authors (and their research teams) addressing adolescent HIV prevention or the adolescent HIV continuum of care in South Africa. We uploaded summarized tables from our systematic review, along with links to the full text articles and abstracts, into REDCap for participants to access (16). We invited AHI(SA)$^2$ members to complete a questionnaire assessing the feasibility, efficacy, scalability, and costing of each



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intervention (using four-point Likert scales). We also asked members to assign an overall score (rating of 0 to 100) regarding the potential impact of the intervention on the continuum of care for ALWH in South Africa. We encouraged the participants to follow the GRADE guidelines in their assessment of the quality of the studies; however, participants did not have access to the GRADE table. AHI(SA)$^2$ leadership decided *a priori* that a score of 70 or greater would be a reasonable estimate of clinically meaningful effects and would be required to be included in the second-round analysis.

All AHI(SA)$^2$ members (n=42) received a secure link to the REDCap survey via email to complete the first-round questionnaire focused on intervention efficacy (August 24, 2020 – September 23, 2020). Results were discussed at the first AHI(SA)$^2$ virtual meeting on September 25, 2020 with the 25 participants who attended. Interventions with a prespecified median overall impact score less than 70 or with an average efficacy score less than three (out of four on the Likert scale) were removed from the next round questionnaire. In preparation for the second round, we updated the literature search from July 31, 2020 to September 30, 2020 and invited AHI(SA)$^2$ members to nominate additional studies for inclusion. The REDCap questionnaire was updated and emailed to AHI(SA)$^2$ members who participated in the first-round questionnaire (n=29) to evaluate feasibility and scalability (September 25, 2020 to December 15, 2020). We sought to formulate an expert consensus on the interventions for ALWH with highest potential be taken to scale and to highlight areas in which future interventions are needed (17, 18).

*Patient and public involvement*

Patients or the public were not involved in the design or conduct of this study. The research questions were formulated by the AHI(SA)$^2$ leadership team. Results of this analysis were presented at regular AHI(SA)$^2$ (www.ahisa2.org) meetings.

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## Results

*Systematic review*

Our initial search yielded 273 relevant articles. Of these, 110 were removed due to duplicate publications or interventions, and 112 did not include interventions addressing the adolescent HIV continuum of care in South Africa, leaving 25 full text articles for review. We excluded 16 of the 25 studies because nine lacked an intervention, three focused on HIV primary prevention, and four met one of the following exclusionary criteria: literature review only (n=1), no primary results (n=1), not focused on adolescents (n=1), or not conducted in South Africa (n=1). We did not find any additional unpublished studies in the review of conference abstracts. Overall, nine published papers with sufficient data were included in the first round of the modified Delphi analysis addressing the adolescent and young adult HIV continuum of care in South Africa.

*Modified Delphi analysis*

Participants: AHI(SA)[2] consisted of 42 eligible individuals involved in adolescent HIV research in South Africa. For the first round of the modified Delphi analysis, 29 (69%) participated in the REDCap survey. Participant characteristics are presented in **Table 1**. Of the 29 who initiated the survey, 21 (72%) completed it. Of the 29 participants who were sent the second-round survey, 21 (72%) initiated it and 18 (62%) completed the survey.

**Table 1: Participant characteristics in two rounds of the modified Delphi Analysis.**

Publications:

Among the nine initial published articles containing interventions addressing the adolescent HIV continuum of care in South Africa, five interventions focused on HIV diagnosis, two on ART adherence, and two on retention in care as indicated in **Supplement 4**. No studies addressed linkage to care or transition from pediatric to adult care. For study design, five studies were randomized



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controlled trials, three were retrospective cohort analyses, and one was a pre-post evaluation. Two studies of the nine discussed the cost of the intervention.

Intervention evaluation:

Below we report summaries of the individual published interventions for ALWH in South Africa separated by steps in the continuum of care. Within each step, the interventions are listed beginning with the highest overall median impact rating in the modified Delphi Analysis (**Supplement 4 and Table 2**).

**Supplement 4:** Interventions to improve the continuum of care for adolescents and young adults living with HIV in South Africa

**Table 2:** Second round modified Delphi analysis of interventions to improve the continuum of care for adolescents living with HIV in South Africa

HIV Diagnosis (n=5)

Five interventions addressed HIV diagnosis through expanded HIV testing among adolescents. Four of these were randomized controlled trials (19-22) and one was a pre-post evaluation (23). Three of these studies met criteria for inclusion in both rounds of the modified Delphi analysis (19, 20, 22). Two of the three studies showed high rates of HIV testing uptake and acceptability of in-home HIV testing and one showed strong short-term effects.

Shanaube et al. reported on the 11,175 adolescents randomized to receive the full intervention of door-to-door HIV testing and counseling with support for linkage to care delivered by community health workers compared to standard care in the Zambia PopART for Youth study (22). HIV testing was done for 80% of participants with awareness of a positive HIV test increased by a three-fold by the third round of interventions and a median time to ART linkage of five months; however, there was no reported comparison to standard of care for HIV testing. Doherty et al. performed a cluster randomized clinic trial among adolescents evaluating door-to-door home-based HIV testing



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performed by lay counselors compared to standard of care (19). HIV testing was significantly higher among in adolescents receiving home-based testing (69%) compared to standard of care (47%) (prevalence ratio 1.54, 95% CI 1.32 to 1.81). Pettifor et al. showed high uptake of HIV self-testing among young women. This study randomized 287 young women ages 18 to 26 to choose between HIV self-testing kits or receive clinic-based HIV testing compared to HIV testing in government clinics (20). In the choice arm, 95% chose self-testing. After three months, 92% of the choice arm and 43% of the standard of care arm reported HIV testing a difference of 49%. By nine months, the difference reduced to 25%.

The remaining two studies did not meet the threshold to continue to the second round of the modified Delphi analysis due to lack of demonstrated efficacy as determined by the first round of the modified Delphi analysis; both used soccer as a means to increase HIV testing. Hershow et al. conducted a pre-post analysis of the SKILLZ Street intervention by Grassroot Soccer where female coaches delivered afterschool education program consisting of 10 two-hour bi-weekly sessions with 1,953 female adolescents ages 12 to 16 years (23). Among the participants, 69% tested for HIV by the end of the intervention. There was no control group or comparison to baseline testing. Rotheram-Borus et al. randomized 142 male adolescents between the ages of 18 and 25 to immediately participate in Grassroot Soccer with trained coaches, rapid HIV testing, and vocational training compared to delayed participation in the control group (21). There was no significant difference in HIV testing among the immediate group (29%) compared to the delayed uptake group (24%) after the intervention.

ART adherence (n = 2)

Two studies evaluated interventions designed to improve adolescent ART adherence (24, 25). Both were included in first round of the modified Delphi analysis, but only one continued to the second round due to lack of perceived efficacy. Cluver et al. evaluated the provision of social protections



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on adherence among 1,059 adolescents ages 10 to 19 years in an observational cohort (25). They found that food provision (adjusted odds ratio [aOR] 0.57, 95% CI 0.42 – 0.76), attendance at HIV support groups (aOR 0.60, 95% CI 0.40 – 0.91) and parental supervision (aOR 0.46, 95% CI 0.43 – 0.73) were associated with lower rates of self-reported non -adherence to ART in the prior week. Bhana et al.'s VUKA Family Program was not included in the second round of the modified Delphi analysis due to lack of clear efficacy. The intervention consisted of 10 sessions over three months and used a culturally tailored cartoon storyline of step-by-step guidance for problem solving skills and facilitated family discussions (24). ART adherence (last time missed medication) increased over time from baseline to completion of the study, but there was no comparison to standard of care.

Retention in care (n=2)

Two studies were designed to improve retention in care, and both were included in the first and second rounds of the modified Delphi Analysis. Zanoni et al. evaluated a weekend-based, adolescent-focused clinic that offered peer support, group activities, counseling, and meals compared to standard of care in a retrospective cohort analysis (26). Retention in care and viral suppression were higher among adolescents who attended the adolescent-friendly clinic (95% and 91%, respectively) compared to standard of care (85% and 80%, respectively; aOR 3.5, 95% CI 1.2 – 11.1; p=0.018 and aOR 2.5, 95% CI 1.1 – 5.8; p=0.028, respectively). The second study by Fatti et al. evaluated community-based support that included home-based ART education, psychosocial support, opportunistic infection screening, and access to government grants compared to standard of care among 6,706 adolescents aged 10 to 24 years (27). Cumulative loss to follow up was lower in the intervention group (30%) compared to standard of care (39%; aHR = 0.60; 95% CI 0.51 – 0.71; p <0.0001). Although there was no statistically significant difference in viral failure after three or five years; 8.8% in the intervention group had viral failure compared to 37.2% in the standard of care group (aOR 0.24, 95%CI 0.06 – 1.03), the difference may be clinically meaningful.

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**Discussion:**

In our systematic review and modified Delphi analysis of interventions addressing the adolescent HIV continuum of care, we found five feasible and effective interventions with high potential for scale-up to improve adolescent HIV outcomes in South Africa. To improve HIV diagnosis among ALWH in South Africa, in-home HIV testing (20) and HIV self-testing (22) were promising interventions. However, there was a lack of demonstrated efficacy in targeting successful linkage to care after HIV testing. Adherence to ART among adolescents improved with the provision of food support, attendance in HIV support groups, and parental supervision (25). Two interventions improved retention in care: one focused on adolescent-friendly services by providing an after-school hours clinic with peer support, counseling, and meals (26) and the other by providing in-home support for HIV education, psychosocial support, and assistance with applications to governmental grants (27). Unfortunately, no interventions addressed transition from pediatric to adult based care for ALWH in South Africa.

Although adolescents and young adults have the highest incidence of new HIV infections in South Africa, they often do not present to health centers for HIV testing. Structural barriers such as school, transportation, or finances; psychosocial barriers such as the feelings of invincibility; or clinical factors such as the lack of clinical symptoms all contribute to poor engagement in care (1, 6). Therefore, traditional methods including clinic-based opt-out testing are not ideal for this population. Non-traditional methods of intensified case finding are required to identify adolescents living with undiagnosed HIV. In-home HIV testing provided by lay counselors offers a solution to the issues of adolescents not attending health centers. Although this process can be labor intensive and costly, it also offers a mechanism to improve linkage to care once adolescents are diagnosed with HIV. Another potential solution to low HIV testing rates among adolescents and young adults is self-testing which has shown acceptability and uptake but requires interventions to provide counseling to newly diagnosed individuals and facilitate linkage to care.



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Adolescents living with HIV commonly suffer poor adherence to ART (28, 29). Direct interventions addressing adherence among adolescents living with HIV are often require considerable effort and resources and have variable efficacy (30, 31). Indirect measures that improve food and economic security, psychosocial support, and supervision appear to improve adherence among adolescents (25). Future interventions to address adherence should include economic and psychosocial support. In addition, addressing barriers such as depression and substance use which has improved ART adherence in adults may also benefit adolescents and young adults (32).

After-school hours clinics, peer support, and improved connection to clinical staff have been shown to improve retention in care among adolescents living with HIV (33). As recommended by the World Health Organization, the provision of adolescent-friendly services and reimagining of clinical care delivery for adolescents and young adults can strengthen retention (34). For example, many of these services can be provided outside of clinic space and time. In-home or virtual clinical services could be expanded to improve adolescent retention in care.

Interventions require resources in time, personnel, and funding but only two interventions in this review presented cost analyses. For implementors and policymakers to adequately assess intervention scale-up, costing and cost-effectiveness analyses are required. Future interventions addressing the adolescent HIV continuum of care should evaluate cost.

We identified several interventions that could be scaled to improve the adolescent HIV continuum of care in South Africa; however, several gaps remain. Improving HIV diagnosis is critical to improve the continuum of care, but without linkage to care, adolescents cannot access ART and achieve viral suppression. Interventions to improve adolescent HIV testing should include linkage to care, and adolescents require additional support to achieve and maintain viral suppression.



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Moreover, children initiating ART under the care of pediatricians will eventually require transition to adult care; however, we did not find any published interventions addressing transition care for ALWH in South Africa highlighting a key area for future research.

Future interventions addressing the adolescent HIV continuum of care should build on the promising results from the highlighted studies and perhaps include a combination of the most effective strategies to optimize outcomes. Additionally, acceptability of the interventions by adolescents is critical for continued success of potential interventions. (35) Using participant-centered design and direct input from adolescent early in intervention development can likely increase acceptability and improve likelihood of intervention success (36).

This systematic review and modified Delphi analysis has several limitations. First, we only identified interventions that were published in the literature and did not include studies that were in progress or planned. In addition, the modified Delphi analysis did not include researchers who were not yet published, received non-NIH funding, or did not respond to email invitations. However, the AHI(SA)$^2$ network includes highly experienced experts with key knowledge to evaluate interventions in South Africa. In addition, not all AHI(SA)$^2$ members who were invited to participate completed the survey. We experienced drop-out rates of 28% and 38% in rounds one and two of the modified Delphi analysis due to strict completion deadlines prior to scheduled AHI(SA)$^2$ meetings. Although participants were encouraged to consider the GRADE guidelines in considering the quality of the studies, they did not have access to the final GRADE table. However, this review highlights published interventions with quantifiable results addressing the continuum of care for adolescents and young adults evaluated by scholars with interest in this population.

**Conclusion:**



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No single intervention exists to improve each step of the continuum of care for ALWH in South Africa. However, we highlight several promising interventions, specifically in-home HIV testing and self-testing to increase the proportion of ALWH who know their diagnosis and economic support and food support to increase ART adherence. These interventions should be further evaluated as they are implemented to improve adolescent outcomes in the HIV care cascade. Future interventions should work comprehensively across the adolescent HIV continuum of care and be tailored to the specific needs of adolescents.

**Declarations:**

**Competing Interests**: The authors have no conflicts of interest relevant to this article to disclose.

**Author's Contributions:**

Dr. Zanoni conceptualized and designed the study, performed the literature review, conducted the analysis, drafted the initial manuscript, reviewed and revised the manuscript and approved the final manuscript as submitted.

Ms. Ramos performed the literature review, assisted with the analysis, assisted in the drafting of the initial manuscript, reviewed and revised the manuscript and approved the final manuscript as submitted.

Dr. Archary assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Ms. Sibaya assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Donenberg assisted with the conceptualization and design of the study, contributed to the analysis plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Shahmanesh assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Celum assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Pettifor assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Bekker assisted with the conceptualization and design of the study, reviewed and revised the manuscript, and approved the final manuscript as submitted.

Dr. Haberer assisted with the conceptualization and design of the study, contributed to the analysis



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plan, reviewed and revised the manuscript, and approved the final manuscript as submitted.

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**Ethics:** The protocol was evaluated by the Emory University Institutional Research Ethics Board and given a waiver of formal ethics review.

**Availability of data:** The datasets used during the current study are available from the corresponding author on reasonable request.

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Supplement 1: Search terms used for systematic review:

Pubmed search strategy:

((adolescent[MeSH Terms]) OR (teen*) OR (youth*) OR (adoles*) AND ((HIV infection[MeSH Terms]) OR (acquired immune deficiency syndrome[MeSH Terms]) OR (HIV[TIAB]) OR ("AIDS"[TIAB]) OR ("acquired immune deficiency syndrome"[TIAB]) OR ("acquired immunodeficiency syndrome"[TIAB])) AND ("south africa"[TIAB]) AND ((intervention[TIAB])

Google scholar search strategy:
"adolescen* HIV intervention South Africa"

ScienceDierect search strategy:
Key words: “Adolescent HIV intervention South Africa”

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# Supplement 2 | PRISMA Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a literature review. | Title page 1 |
| **ABSTRACT** | | | |
| Abstract | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings.<br>See the PRISMA 2020 for Abstracts checklist for the complete list. | Abstract page 1 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge, i.e., what is already known about your topic. | Page 2, lines 23 - 29 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | Pages 2 – 3, lines 25 - 29 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses with study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | Page 3, lines 33 - 46 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | Page 3, lines 33 - 39 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | Appendix 1 |
| Selection process | 8 | State the process for selecting studies (i.e., screening, eligibility).<br>Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | Pages 3 – 4, lines 48 - 59 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | Page 4, lines 61 - 66 |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | Pages 5 – 6, lines 99 - 107 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were | N/A |

The checklist has been adapted for KIN 4400 Independent Research Study in Kinesiology at the University of Guelph-Humber. Last updated: Dec 9, 2021

*Adapted From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71 *Adapted From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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## Supplement 2 | PRISMA Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| | | excluded. | |
| Study characteristics | 17 | Cite each included study and present its characteristics (e.g., study size, PICOS, follow-up period). | Table 1 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | Page 4, lines 61 - 66 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Table 1 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | Pages 9 - 12 |
| | 23b | Discuss any limitations of the evidence included in the review. | Page 12, lines 267 - 277 |
| | 23c | Discuss any limitations of the review processes used. | Page 12, lines 267 - 277 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | Page 12, lins 280 - 286 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | N/A |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | Not prepared |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | None |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | Supplementary information |
| Competing interests | 26 | Declare any competing interests of review authors. | Supplementary information |
| Availability of data, code, and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | Supplementary information |

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## Supplement 3: Evaluation of the quality of evidence using the GRADE guidelines

| STUDY | DESIGN | QUALITY | CONSISTENCY | DIRECTNESS | MODIFYING FACTORS | POPULATION |
|---|---|---|---|---|---|---|
| **HIV TESTING** | | | | | | |
| **DOHERTY, ET AL.** | Randomized controlled trial | No serious limitation | No important inconsistency | Direct | Cluster randomized | 4,154 adolescent and adults |
| **ROTHERAM-BORRUS, ET AL** | Randomized controlled trial | No serious limitation | No important inconsistency | No effect of intervention | Cluster randomized 2 sites only | 142 adolescent males |
| **HERSHOW, ET AL.** | Observational | No control group | No important inconsistency | Direct without comparison | None | 1,953 adolescent females |
| **SHANAUBE, ET AL.** | Randomized clinical trial | Single arm reported | No important inconsistency | Direct without comparison | None | 11,175 adolescents |
| **PETTIFOR, ET AL.** | Randomized controlled trial | No serious limitation | No important inconsistency | Direct | Decreased effect over time | 287 young women |
| **ADHERENCE** | | | | | | |
| **BHANA, ET AL.** | Randomized controlled trial | No serious limitation | No important inconsistency | Indirect measure of adherence | Self-reported adherence | 65 adolescents |
| **CLUVER, ET AL.** | Observational | No serious limitation | No important inconsistency | Indirect measure of adherence | Self-reported adherence | 1,059 adolescents |
| **RETENTION IN CARE** | | | | | | |
| **FATTI, ET AL.** | Observational | No serious limitations | No important inconsistency | Direct | None | 6,706 adolescents |
| **ZANONI, ET AL.** | Observational | No serious limitation | No important inconsistency | Direct | Large effect size Propensity score matching | 254 adolescents |

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**Supplement 4:** Interventions to improve the continuum of care for adolescents and young adults living with HIV in South Africa

| Authors | Title | Publication and Date | Location | Number of Subjects | Ages (Target Population) | Study Type | Intervention Description | Overall Results | Median Overall Score Round 1 |
|---|---|---|---|---|---|---|---|---|---|
| **Adolescent HIV Testing** | | | | | | | | | |
| Doherty, et al. | Effect of Home-Based HIV Counseling and Testing Intervention in Rural South Africa: Cluster Randomised Trial | *BMJ* 2013 | uMzimkhulu subdistrict, KwaZulu-Natal, South Africa | 4154 Adolescents and Adults | 14-17, 18+ Adult Guardians | Cluster Randomized Controlled Trial; Door-to-Door Testing | Door-to-door home-based HIV testing (HBT) performed by lay counselors compared to standard of care HIV counseling and testing at local clinics. | Testing completed in 69% of HBT compared to 47% in control (PR 1.54 95% CI 1.32 to 1.81) | 72.5 |
| Rotheram-Borus, et al. | Feasibility of Using Soccer and Job Training to Prevent Drug Abuse and HIV | *AIDS and Behavior* 2016 | South Africa | 142 Adolescents | 18-25 Unemployed Males | Community Randomized Controlled Trial | Randomized neighborhoods to immediate intervention or delayed control; Intervention: Grassroot Soccer program with trained coaches, random rapid diagnostic tests for alcohol/drugs, and vocational training. | 29% testing in immediate vs. 24% in delayed (no significant difference) | 50 |
| Hershow, et al. | Using Soccer to Build Confidence and Increase HCT Uptake Among Adolescent Girls | *Sport in Society* 2015 | South Africa | 1953 Adolescents | 12-16 (Females Only) | Pre-Post Evaluation | Survey before and after participation in SKILLZ Street intervention developed by Grassroot Soccer (GRS). SKILLZ Street female coaches delivered an afterschool education program consisting of 10, two hour biweekly sessions. | 69% tested for HIV<br>No control group. | 63 |
| Shanaube, et al. | Community Intervention Improves Knowledge of HIV Status of Adolescents in Zambia: Findings from HPTN 071-PopART for Youth Study | *AIDS* 2017 | Results from Zambia only; Intervention also conducted in-South Africa | 11 175 Adolescents | 15-19 | Community-Randomized Clincal Trial | **HPTN 071 (PopART)**<br>Door-to-door HIV prevention package delivered by community health worders. Intervention included in-home HIV Counseling and Testing (HCT), support for linkage to care for individuals identified living with HIV, ART adherence support, referrals for voluntary medical male circumcision | HCT uptake 80% (8,707/10,809) and awareness of an HIV-positive status tripled from 75 to 210. Home visits from providers saw the percentage of ALWH aware of their HIV status rise from 27.6% to 88.5%.<br><br>No comparison with standard care. | 76 |

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| Pettifor, et al | HIVself-testing among young women in rural South Africa: A randomized controlled trial comparing clinic-based HIVtesting to the choice of eitherclinict esting or HIVself-testing with secondary distribution to peers andartners | *EClinical Medicine* 2020 | Mpumlanga, SouthAfrica | 287 women | 18 - 26 | Randomized Cotrolled Trial | Standard of care HIV counseling and testing (HCT) at government clinics vs. choice of clinic based HCT or HIV self-testing kits.<br><br>+referrals for peers to recelive clinic based HCT or HIV self-testing kits | 146 randomized to HCT and 141 to choice. In the choice arm 95% (n=135) chose HIV self testing kits. After 3 months a significantly higher percent (92%) of participants in the choice arm reported HIV testing compared to (43%) the HCT arm with a difference of 43% (95% CI 40% - 58%). By 9 months the difference decreased to 25% (95% CI 17% - 33%). Participants in the choice arm were more likely to invite peers or partners to test (94% vs. 76%) | 70 |
| **Adherence** | | | | | | | | | |
| Bhana, et al. | The VUKA Family Program: Piloting a Family-Based Psychosocial Intervention to Promote Health and Mental Health Among HIV Infected Early Adolescents in South Africa | *AIDS Care* Jan 2014 | KwaZulu-Natal, South Africa | 65 Adolescents and Their Family Members | 10-13 | Randomized Controlled Trial | **VUKA Family Program** Inspired by CHAMP, the intervention consisted of 3 months and ten sessions involving a culturally-tailored cartoon storyline and curriculum. The curriculum follows the story and provdes step-by-step guidance for problem solving and facilitating discussions. ALWH were assessed for feelings of stigma, knowledge of ART, adherence depression and caregiver support. | ART adherence: Intervention (I) 1.10, Control (C) -. 43 at baseline; Follow-up, $p < .05$.<br><br>Caregiver/child communication: (I) .41, (C) -.07, $p < .09$.<br><br>Youth knowledge of HIV treatment: (I) .27; (C) -.74, $p < .08$. | 68.5 |
| Cluver, et al. | Achieving Equity in HIV-Treatment Outcomes: Can Social Protection Improve Adolescent ART-Adherence in South Africa? | *AIDS Care* 2016 | Eastern Cape, South Africa | 1059 Adolescents | 10-19 | Observational Cohort | Evaluated the provision of social protections on adherence. Social provisions included government cash transfers, food security, school fees/materials, school feeding, and clothing; HIV support group, sports groups, choir/art groups, positive parenting and parental supervision/monitoring. | Three social protection provisions were associated with reduced non-adherence:<br><br>Food provision (aOR .57, CI .42–.76, p < .001);<br><br>HIV support group attendance (aOR .60, CI .40–.91, p < .02); High parental/caregiver supervision (aOR .56, CI .43–.73, p < .001). | 75 |
| **Retention in Care** | | | | | | | | | |

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| Fatti, et al. | The Effectiveness and Cost-Effectiveness of Community-Based Support for Adolescents Receiving Antiretroviral Treatment: An Operational Research Study in South Africa | *Journal of the International AIDS Society* 2018 | KwaZulu-Natal/Western Cape/Eastern Cape/Mpumalanga, South Africa | 6706 Adolescents | 10-24 | Retrospective Cohort Study | Community based support (CBS) included home-based ART-related education, psychosocial support, symptom screening for opportunistic infections and support to access government grants. CBS was compared to standard of care. | Cumulative LTFU was 40% lower amongst participants receiving CBS (29.9%) compared to participants without CBS (38.9%), aHR = 0.60 (95% CI: 0.51 to 0.71); p < 0.0001).<br><br>Viral failure: at 3 years – no difference; at five years 18.8% CBS participants versus 37.2% non-CBS participants had viral failure, adjusted odds ratio = 0.24 (95% CI: 0.06 to 1.03). | 80 |
| Zanoni, et al. | Higher Retention and Viral Suppression with Adolescent-Focused HIV Clinic in South Africa | *PLoS One* 2017 | KwaZulu-Natal, South Africa | 254 HIV+ Adolescents | 13-24 | Retrospective Cohort Analysis | Weekend adolescent-focused clinic with peer support, group activities, counseling, and meals compared to standard of care. | Retention: 95% in adolescent-friendly clinic vs. 85% in standard clinic (OR 3.5; 95% CI 1.2 – 11.1; p=0.018).<br><br>Viral Suppression: 91% adolescent-friendly clinic vs. 80% in standard clinic. (OR 2.5; 95% CI 1.1 – 5.8; p=0.028 | 88 |

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